

SQL - ACID Properties

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- A → Atomicity → A transaction either should happen successfully or shouldn't happen at all.
- C → Consistency → Database should remain constant before & after the transaction
- I → Isolation → When multiple transactions occur on a table, each transaction should execute independently & should not interfere with others.
- D → Durability → Objects/transaction should be available. Even after system failure.

What is a table?

A table is an object formed with named columns & data types to store the information of rows

Col 1	Col 2	Col 3	Col 4	Col 5

Columns are formed vertically

Rows are formed horizontally

Types of Data types (MySQL)

* String → char, varchar,

* Numeric → Tinyint, Smallint, Bigint, Float, double, decimal

* Date → Date, Time, Timestamp, Datetime

- * Date → Date, Time, Timestamp, Datetime
- * Binary → Blobchar

Char_col → char(10) → 10 length character value.

Varchar → variable length of characters

(255) 1, 2

Create Table <table_name> (

Column name ←	Col-1	char(5),	} Data types
	Col2	varchar(10),	
	Col3	smallint,	
	Col4	int,	
	Col5	float,	
	Col6	date / datetime / timestamp	

);

Types of SQL Commands

- * DDL → Data Definition Language
- * DML → Data Manipulation Language
- * DQL → Data Query Language
- * DCL → Data Control Language
- * TCL → Transaction Control Language.